

Date	31 January 2025	Maximum Marks	4 Marks
Team ID	SWTID-2026-5019	Project Name	Import Data using Transform Maps (Spreadsheet)

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Requirements Engineering Specification
ServiceNow ETL & Integration Track

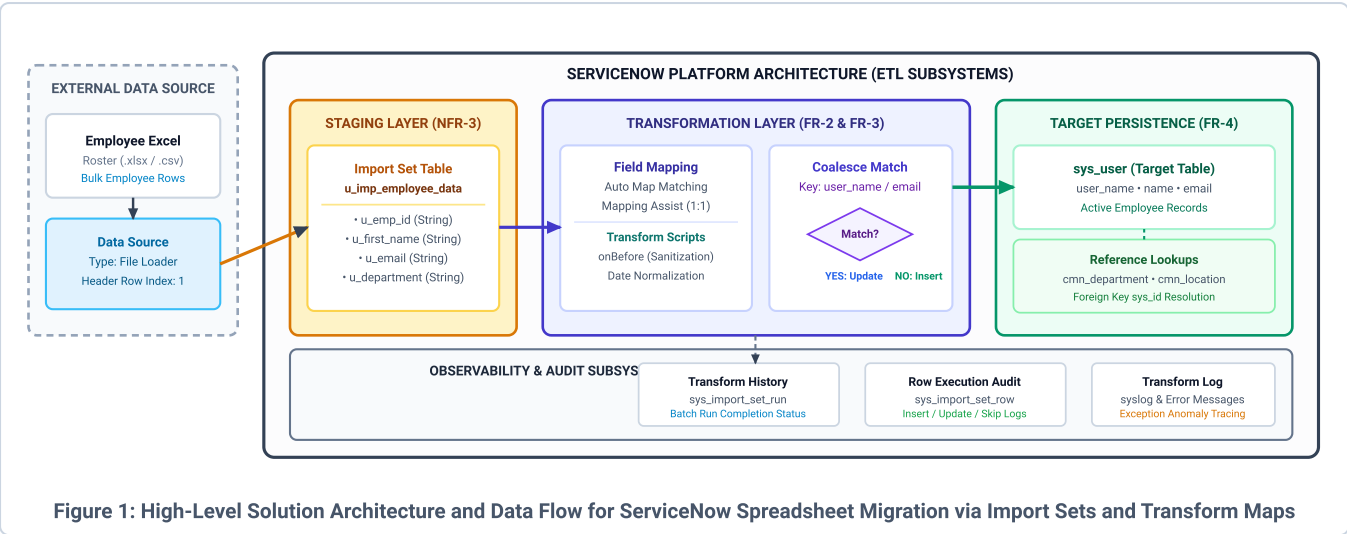
PROJECT OVERVIEW & CONTEXT

This micro project focuses on importing structured data from an external spreadsheet into the ServiceNow platform using **Import Sets** and **Transform Maps**. The project simulates a real-world scenario where bulk employee data is received in Excel format and must be migrated into ServiceNow efficiently and accurately. The spreadsheet data is loaded into an Import Set table (staging buffer), and a Transform Map is configured to map source fields to the target table (sys_user), execute coalesce deduplication, and validate records to ensure data integrity.

Functional Requirements		Operational Capabilities of Proposed Solution
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Staging Isolation	- FR-1.1: Ingest bulk employee Excel files (.xlsx / .csv) via native ServiceNow Data Source loaders. - FR-1.2: Dynamically generate an isolated Import Set staging table (u_imp_employee_data) to quarantine raw data. - FR-1.3: Parse spreadsheet header indices and validate loaded staging row counts against source records.
FR-2	Schema Field Mapping	- FR-2.1: Configure a dedicated Transform Map establishing linkage between staging and target sys_user table. - FR-2.2: Auto-map direct 1:1 matching fields (First Name, Last Name, Email, Phone) via Auto Map Matching Fields. - FR-2.3: Map custom columns manually using Mapping Assist (e.g., source u_emp_id to target user_name).
FR-3	Identity Deduplication (Coalesce)	- FR-3.1: Designate unique identifier (User ID / Email) with Coalesce = true to prevent duplicate accounts. - FR-3.2: Automatically execute updates on matching existing user records without overwriting unmapped fields. - FR-3.3: Automatically insert new user records when no existing coalesce match is found in the target table.
FR-4	Data Transformation & Validation	- FR-4.1: Execute onBefore scripts to normalize disparate date formats into standard YYYY-MM-DD. - FR-4.2: Resolve reference lookups (Department cmn_department, Location cmn_location) to prevent invalid references. - FR-4.3: Generate execution telemetry in Transform History (sys_import_set_run) tracking inserts, updates, and errors.

Non-functional Requirements		Quality Attributes, Security & Architectural Constraints
FR No.	Non-Functional Requirement	Description & Architectural Quality Benchmark
NFR-1	Usability	Intuitive visual mapping via ServiceNow Mapping Assist studio; requires zero complex database coding for standard column assignments, enabling junior administrators to complete migrations with minimal training.
NFR-2	Security	Enforce strict Role-Based Access Control (RBAC) restricting import executions to authorized roles (admin, import_admin); ensure staging data encryption at rest and in transit without exposing sensitive credentials.
NFR-3	Reliability	Guaranteed isolation of staging buffer prevents malformed spreadsheet rows from corrupting production tables; provides automatic transaction rollbacks and row-level error handling (ignore = true) for bad data.
NFR-4	Performance	High-speed ETL processing capable of transforming 500+ spreadsheet rows in under 60 seconds; indexed coalesce queries ensure zero table locking or latency impact on live platform operations.
NFR-5	Availability	Native integration with enterprise high-availability ServiceNow cloud infrastructure (99.99% uptime); imports can execute asynchronously via background workers during non-peak business hours.

Architectural Purpose: The solution architecture operationalizes the functional (FR-1–FR-4) and non-functional (NFR-1–NFR-6) requirements through a modular multi-tier ETL pipeline. Uncleaned Excel spreadsheets are staged in an isolated buffer table, sanitized and mapped through the transformation engine, validated against coalesce rules, and committed to target enterprise tables with complete audit logging.



Architectural Subsystem	ServiceNow Module	Requirement Realized	Technical Fulfillment & Verification Mechanism
1. Ingestion Layer	Data Source (File Loader)	FR-1, NFR-4	Ingests bulk Excel sheets; parses column headers; enforces file structure indexing with sub-minute ingestion throughput.
2. Staging Buffer	Import Set Table (u_imp_*)	FR-1.2, NFR-3	Quarantines raw unvalidated spreadsheet rows; prevents unverified external records from directly contacting live production schemas.
3. Transform Engine	Transform Map & Coalesce	FR-2, FR-3, NFR-1	Provides codeless 1:1 visual mapping and evaluates coalesce matching on User ID to insert new profiles and update existing accounts without duplication.
4. Transformation Logic	onBefore Scripts & Choice Ops	FR-4.1, FR-4.2	Sanitizes disparate date strings, normalizes phone text, and resolves Department reference foreign keys via target sys_id matching.
5. Observability & Target	Target sys_user & History Logs	FR-4.3, NFR-2, NFR-5	Commits authenticated records to active directories under RBAC controls and generates row-level audit telemetry in sys_import_set_run.